60-Minute Interactive Workshop

Hands-On Developer Session

ZERO TO FIRST PUSH

Mastering Git & GitHub: Command Line Fundamentals

Open an account, initialize local repositories, and execute your first terminal push in under one hour.

SPEAKER SCRIPT & WELCOME (MIN 0:00 - 0:02)

"Welcome everyone! By the time you leave this room in 60 minutes, you won't just understand GitHub conceptually — you will have created an account, authored code locally, and pushed your first live commit directly through your terminal."

Workshop Agenda & Roadmap

4 Core Milestones

00:00 – 00:12

1. Mental Models

Git vs GitHub demystified & account verification.

00:12 – 00:25

2. Authentication

Configuring Git user credentials & Personal Access Tokens (PAT).

00:25 – 00:50

3. The CLI Workflow

Init, Stage, Commit, Link, and Push live to remote repository.

00:50 – 01:00

4. Verification & QA

Inspecting GitHub web UI & developer portfolio fundamentals.

SPEAKER SCRIPT (MIN 0:02 – 0:05)

"Here is our path for today. Notice that the largest block of time is dedicated to hands-on command-line mechanics. Don't worry if you've never touched a terminal before — we will move step by step."

The Core Concept: Git vs. GitHub

Conceptual Clarity

Git (Local Engine)

Video game **"Save Point" system** on your computer. Tracks file history, revisions, and snapshots locally without internet.

↔

GitHub (Cloud Host)

"Google Drive for Developers" in the cloud. Stores your Git save points online so teams can collaborate and review code.

SPEAKER SCRIPT & METAPHOR (MIN 0:05 – 0:09)

"Think of Git as your local camera capturing snapshots of your project on your hard drive. GitHub is Instagram — the cloud platform where you upload those snapshots to share with the world."

Step 1: Environment Readiness Check

Live Action

1. Test Local Git Installation

Open Command Prompt (Windows) or Terminal (macOS/Linux) and run:

```
git --version
```

Target Response: git version 2.x.x

2. Account Creation Action

- [x] Navigate to **github.com/signup**
- [x] Sign up using your primary email
- [x] Verify account via email link

TROUBLESHOOTING SCRIPT & HELPER CALL (MIN 0:09 – 0:15)

"Everyone run `git --version` right now. If you get an error saying 'command not found', please raise your hand immediately — our workshop assistants will help you launch Git Bash."

Step 2: Identity Configuration

One-Time Setup

Git tags every snapshot with your signature. Run these two commands in your terminal:

```
git config --global user.name "Your Name"
git config --global user.email "your_email@example.com"
```

⚠ CRITICAL: Use the exact same email address you used to register your GitHub account!

SPEAKER SCRIPT (MIN 0:15 – 0:20)

"This registration step ensures GitHub correctly links your local commits to your online profile avatar and contribution activity graph."

Step 3: Terminal Authentication

Key Security Insight

Standard Security Requirement

GitHub **does not accept account passwords** in the command line interface.

Selected Method: GitHub CLI

Run `gh auth login` or use Personal Access Tokens (PAT) generated under Settings.

```
gh auth login
```

SPEAKER SCRIPT & PITFALL WARNING (MIN 0:20 – 0:25)

"This is where 90% of beginners get stuck. When terminal prompts for a password, your web account password will fail. We use `gh auth login` for browser authentication, or generate a Personal Access Token."

Step 4: Create Remote Repository

GitHub Web UI

1. Click the green **"New"** repository button on GitHub.
2. Repository Name: `my-first-repo`
3. Set Visibility: **Public**
4. **UNCHECK** "Add a README file" (we build locally!).
5. Click **"Create repository"**.

TARGET URL FORMAT:

```
https://github.com/YOUR-USERNAME/my-first-repo.git
```

SPEAKER SCRIPT (MIN 0:25 – 0:30)

"Keep this GitHub browser tab open after clicking create. GitHub will show you a page with exact commands that we will now run in our command prompt."

Step 5: Local Workspace & File Creation

Terminal Navigation

Execute these three commands to create your project folder and first file:

```
mkdir my-first-repo
cd my-first-repo
echo "# My First GitHub Project" > index.html
```

`mkdir`	= Make Directory
`cd`	= Change Directory
`echo`	= Write text to file

SPEAKER SCRIPT (MIN 0:30 – 0:35)

"We have just created a project folder and generated an `index.html` file inside it entirely through terminal commands. Next, we convert this folder into a Git repository."

Step 6: Initialize & Stage Code

Git Mechanics

```
git init
git status
git add index.html
git status
```

The Staging Area Concept

`git add` moves files to the **Staging Area** (loading items into the shipping box before sealing it).

SPEAKER SCRIPT (MIN 0:35 – 0:40)

"Notice how `git status` changes color from red (untracked) to green (staged). `git status` is your best friend — run it whenever you want to check what Git is thinking."

Step 7: Commit Your Changes

Creating Save Point

Seal the staged changes into a permanent local snapshot with a descriptive message:

```
git commit -m "Create initial index.html file"
```

Good Commit Messages

"Add user login validation"
"Fix navbar spacing on mobile"

Bad Commit Messages

"stuff"
"asdfghjkl / fixed stuff"

SPEAKER SCRIPT (MIN 0:40 – 0:45)

"The `-m` flag stands for message. Always write commit messages in the present tense describing what the commit does. Your future teammates will thank you."

Step 8: Link Remote & Push to Cloud

Final Execution

Connect your local folder to your GitHub URL and push the commits:

```
git branch -M main
git remote add origin https://github.com/YOUR-USERNAME/my-first-repo.git
git push -u origin main
```

SPEAKER SCRIPT & LIVE VICTORY MOMENT (MIN 0:45 – 0:50)

"Press Enter on `git push`! Now go back to your browser, refresh your GitHub repository page, and watch your local code appear online live!"

Mission Accomplished

You are now a GitHub Contributor!

Core 5 Commands

`init, status, add, commit, push`

Developer Portfolio

Your public GitHub profile acts as your live resume.

Next Mastery Skill

Branching (`git branch`) & Pull Requests.

Q&A & WRAP-UP (MIN 0:50 – 1:00)

"Congratulations! Open up for questions now. Direct remaining attendees to keep practicing by modifying their `index.html` file and pushing a second commit!"